

Slide 06b should name the clades and say why each length pattern matters

Community	Arms	Length evidence	Biological story	One-sentence framing
C7 C7 acrocentric p	13p, 14p, 15p, 21p, 22p	n = 763 PHRs median / p90 = 500 / 500 kb max reaches 500 kb	rDNA-adjacent acrocentric p-arms are essentially full-window PHRs; this is the clearest length outlier.	Say: the length plot already knows the acrocentric community.
C14 C14 PAR2	Xq, Yq	n = 431 PHRs median / p90 = 330 / 330 kb max reaches 500 kb	PAR2 sits at the canonical pseudoautosomal q-end scale and gives the audience a familiar length reference.	Say: ordinary subtelomeric PHRs are in the same size regime as PAR2.
C15 C15 PAR1	Xp, Yp	n = 419 PHRs median / p90 = 500 / 500 kb max reaches 500 kb	PAR1 makes the sex-chromosome p-ends look like full-window shared sequence, not a generic subtelomeric tail.	Say: both PAR clades are visible before community detection.
C1 C1 DUX4/D4Z4	4q, 10q	n = 714 PHRs median / p90 = 145 / 215 kb max reaches 500 kb	The 4q/10q DUX4-D4Z4 pair has a moderate center plus a real 500 kb tail, matching copy-number diversity.	Say: the DUX4 clade is not a question mark; it is a named community.
C2 C2 10p-18p	10p, 18p	n = 889 PHRs median / p90 = 70 / 105 kb	The Linardopoulou 10p/18p pair is a community callout more than a long-tail callout: short, tight, and specific.	Say: not every named clade is long; some are specific exchange pairs.
No signal No interchrom PHR	2p, 3p, 5p, 8q, 11q, 14q, 18q	n = 0 in TSV absent from 41-arm matrix	These arms have no non-empty inter-chromosomal PHR intervals in the current TSV and therefore do not enter the 41-arm matrix.	Say: the blank panels are biological absence, not a plotting failure.

Data check: current TSV zero-signal arms are 2p, 3p, 5p, 8q, 11q, 14q, 18q; older notes list six and should be reconciled during final integration.